

Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `Track 2H` quantile/probabilistic campaign into the official `Track 2` offline verification matrix.

Decision

- `Track 2H` quantile/probabilistic regression is verified as an exploratory dispersion-aware baseline.
- No quantile/probabilistic candidate is promoted over the current accepted `Track 2` leaders.
- The project continues to maintain three parallel best surfaces: `Fw` , `Bw` , and `global` .
- Within the quantile/probabilistic package, the strongest forward candidate is `track2h_gaussian_nll_global` .
- Within the quantile/probabilistic package, the strongest backward candidate is `track2h_quantile_p10_p50_p90_Bw` .
- Within the quantile/probabilistic package, the strongest global candidate is `track2h_gaussian_nll_global` .
- The probabilistic package improves the previous `Track 2H` robust-loss package on the best `global` and `Bw` surfaces, but not on the best `Fw` surface.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:
output/validation_checks/track2_reference_comparison/2026-06-12-14-44-45__track2_full_directional_family_matrix_track2h_quantile_probabilistic_track2_refresh_2026_06_12/validation_summary.yaml ;
- per-condition metrics:
output/validation_checks/track2_reference_comparison/2026-06-12-14-44-45__track2_full_directional_family_matrix_track2h_quantile_probabilistic_track2_refresh_2026_06_12/per_condition_metrics.csv ;
- best-model collage report:
doc/reports/analysis/track2/best_model_collage_report/[2026-06-12]/track2_best_model_collage_report.md ;
- multi-model curve comparison report:
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-12]/track2_multi_model_curve_comparison_report.md ;
- operator launch logs:
output/validation_checks/track2_operator_launch_logs/2026-06-12-14-44-41_track2h_quantile_probabilistic_track2_refresh_2026_06_12/ .

Candidate Refresh

The refresh added 6 registry-backed Track 2H quantile/probabilistic candidates covering quantile and Gaussian uncertainty profiles across the global , Fw , and Bw surfaces.

Surface	Profile	Deterministic Curve	Candidate
global	quantile_p10_p50_p90	p50	track2h_quantile_p10_p50_p90_global
Fw	quantile_p10_p50_p90	p50	track2h_quantile_p10_p50_p90_Fw
Bw	quantile_p10_p50_p90	p50	track2h_quantile_p10_p50_p90_Bw
global	gaussian_nll	mu	track2h_gaussian_nll_global
Fw	gaussian_nll	mu	track2h_gaussian_nll_Fw
Bw	gaussian_nll	mu	track2h_gaussian_nll_Bw

The matrix source group is track2h_quantile_probabilistic_registry . The matrix now contains 147 candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
Fw	forward-only training or archive	forward Track 2 curves only
Bw	backward-only training or archive	backward Track 2 curves only
global	forward and backward training together	both directions, reported by direction and combined

The probabilistic heads are not compared as raw multi-channel tensors. The official matrix evaluates only the deterministic TE curve exposed by each checkpoint: `p50` for quantile heads and `mu` for Gaussian NLL heads.

Current Leaders

Overall Direction Leaders

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	<code>rcim_retuned_GBM19_Fw</code>	0.001089	0.001299	2.372	4.912
backward	<code>periodic_gru_sequence_Bw</code>	0.002392	0.002639	5.466	14.820
global forward	<code>periodic_gru_sequence_global</code>	0.002777	0.003025	6.267	13.580
global backward	<code>periodic_gru_sequence_global</code>	0.002630	0.002872	6.010	12.693

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	<code>tree_Bw</code>	0.003258	0.003651	7.051	14.116
global forward	<code>tree_global</code>	0.002998	0.003364	6.590	11.601
global backward	<code>tree_global</code>	0.003290	0.003702	7.118	13.703

Track 2H Quantile Probabilistic Result

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
global forward	<code>track2h_gaussian_nll_global</code>	<code>gaussian_nll</code>	0.002951	0.003246	6.524	15.237
Bw	<code>track2h_quantile_p10_p50_p90_Bw</code>	<code>quantile_p10_p50_p90</code>	0.002935	0.003250	6.307	16.529
Bw	<code>track2h_gaussian_nll_Bw</code>	<code>gaussian_nll</code>	0.003001	0.003303	6.488	14.856
global backward	<code>track2h_gaussian_nll_global</code>	<code>gaussian_nll</code>	0.003068	0.003372	6.627	15.928
Fw	<code>track2h_gaussian_nll</code>	<code>gaussian_nll</code>	0.003156	0.003415	7.008	10.991

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw						
global forward	track2h_quantile_p10_p50_p90_global	quantile_p10_p50_p90	0.003188	0.003469	7.059	12.765
Fw	track2h_quantile_p10_p50_p90_Fw	quantile_p10_p50_p90	0.003276	0.003545	7.279	12.393
global backward	track2h_quantile_p10_p50_p90_global	quantile_p10_p50_p90	0.003563	0.003909	7.816	15.378

The best probabilistic global candidate improves over the best robust-loss global candidate from 0.003401 to 0.003009 combined Track 2 MAE. The best probabilistic Bw candidate improves over the best robust-loss Bw candidate from 0.003078 to 0.002935 Track 2 MAE. The best robust-loss Fw candidate remains slightly better than the best probabilistic Fw candidate: 0.003134 versus 0.003156 Track 2 MAE.

The strongest probabilistic Bw candidate also beats the static tree_Bw baseline on Track 2 MAE, 0.002935 versus 0.003258 , but it remains behind the accepted periodic_gru_sequence_Bw branch.

Visual Evidence

The 2026-06-12 collage and overlay bundles were regenerated after the probabilistic refresh. They now include explicit Track 2H quantile probabilistic sections for Fw , Bw , and global collage evidence, plus explicit Fw and Bw probabilistic overlay sections.

The visual package supports the matrix decision: the probabilistic heads are a useful dispersion-aware ingredient, especially for Bw and the global surface, but they do not replace the strongest periodic temporal branch or the forward paper-reference leader.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-12	Track 2H quantile/probabilistic refresh	6 quantile and Gaussian global , Fw , and Bw candidates	included in the 147 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-06-11	Track 2H robust-loss dispersion-aware refresh	9 robust global , Fw , and Bw candidates	included in the 141 - candidate matrix	dated collage and overlay bundles regenerated with Track 2H sections	verified exploratory baseline; not promoted
2026-06-10	Track 2G curve-aware training refresh	12 pointwise-control, centered-shape, offset, and full-composite global , Fw , and Bw candidates	included in the 132 - candidate matrix	dated collage and overlay bundles regenerated with Track 2G sections	verified exploratory baseline; not promoted
2026-06-08	Track 2F-bis harmonic-offset probe refresh	6 clean and harmonic global , Fw , and Bw candidates plus 3 rechecked Track 2F candidates	included in the 120 - candidate matrix	dated collage and overlay bundles regenerated with Track 2F-bis sections	verified exploratory baseline; not promoted
2026-06-04	Track 2F offset-aware probe refresh	3 global , Fw , and Bw sequential	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
		residual offset candidates			
2026-05-28	Wave 2C residual harmonic temporal hybrid refresh	18 global , Fw , and Bw residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2C sections	verified exploratory baseline; not promoted over Wave 2B or accepted Track 2 baselines
2026-05-26	Wave 2B harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM global , Fw , and Bw candidates	included	collage and overlay reports refreshed	strongest repository-owned neural branch

Closeout Decision

Track 2H quantile/probabilistic regression does not change the accepted Track 2 baseline. The current direction-parallel decision remains:

- Fw : rcim_retuned_GBM19_Fw remains the strongest overall forward candidate; within this package, track2h_gaussian_nll_Fw is the strongest forward-only probabilistic candidate, but track2h_gaussian_nll_global performs better on forward curves.
- Bw : periodic_gru_sequence_Bw remains the strongest repository-owned backward candidate; track2h_quantile_p10_p50_p90_Bw is the strongest probabilistic backward candidate and improves over both the robust-loss Track 2H backward candidate and tree_Bw on Track 2 MAE.
- global : periodic_gru_sequence_global remains the strongest repository-owned bidirectional neural candidate; within the probabilistic package, track2h_gaussian_nll_global is clearly the best global surface and improves over the best robust-loss global Track 2H result. The next modeling step should keep probabilistic losses as a validated ingredient for dispersion-aware training. The evidence supports moving to mixture-density heads and latent-state or hysteresis-aware variants before deciding which mechanisms belong inside the integrated multi-task / multi-head branch.